Application of Information & Communication Technologies

Lecture-17

Recap of Lecture 15 & 16

- Application Software (part 2)
- Presentation
- Multimedia
- Other Applications

Overview of Lecture 17

- What is a Database?
- Data, Information, and Tables
- Components of a Database System (DBMS)
- Database vs File System

Databases

- A collection of related data organized for quick access and retrieval.

- Stores data in tables (rows and columns).

- Example:
  - Student records, bank transactions, library catalog.

Why Databases Are Needed

Problem with Files	Benefit of Databases
Data duplication	Centralized storage
Difficult to search	Easy querying
No security	Controlled access
No relation between files	Data relationships

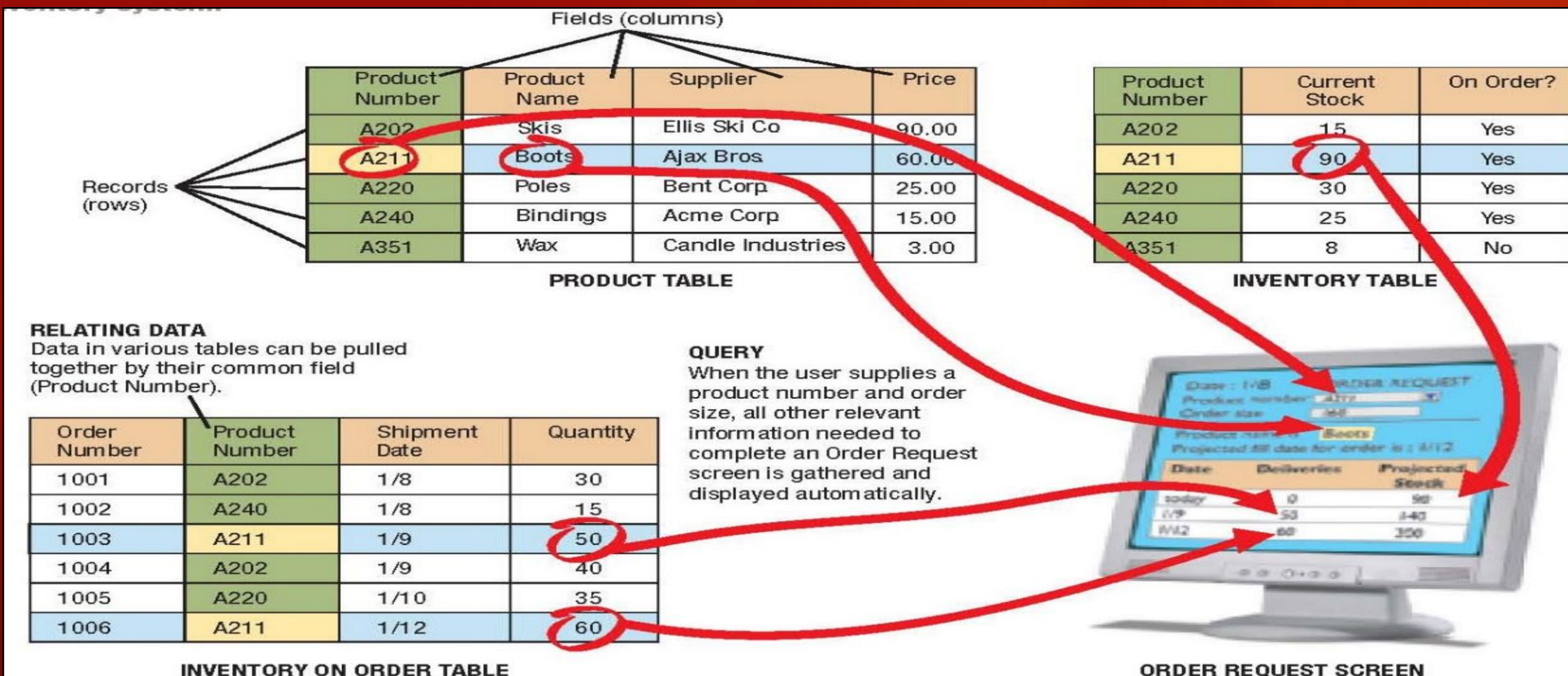
Data, Information, and Tables

- Data: Raw facts (e.g., “Ali”, “21”, “CS”).
- Information: Processed data with meaning (e.g., “Ali is 21 years old”).
- Table: A structured collection of data (rows = records, columns = fields).
  - sample student table

StudentID	Name	Age	Department	GPA
101	Ali	21	CS	3.6
102	Sara	22	AI	3.8
103	Bilal	20	SE	3.4
104	Hina	23	IT	3.9

Relational Database

– **Relational database:** Data from several tables is tied together (related) using a field that the tables have in common.



Product Number	Product Name	Supplier	Price
A202	Skis	Ellis Ski Co	90.00
A211	Boots	Ajax Bros	60.00
A220	Poles	Bent Corp	25.00
A240	Bindings	Acme Corp	15.00
A351	Wax	Candle Industries	3.00

PRODUCT TABLE

Product Number	Current Stock	On Order?
A202	15	Yes
A211	90	Yes
A220	30	Yes
A240	25	Yes
A351	8	No

INVENTORY TABLE

Order Number	Product Number	Shipment Date	Quantity
1001	A202	1/8	30
1002	A240	1/8	15
1003	A211	1/9	50
1004	A202	1/9	40
1005	A220	1/10	35
1006	A211	1/12	60

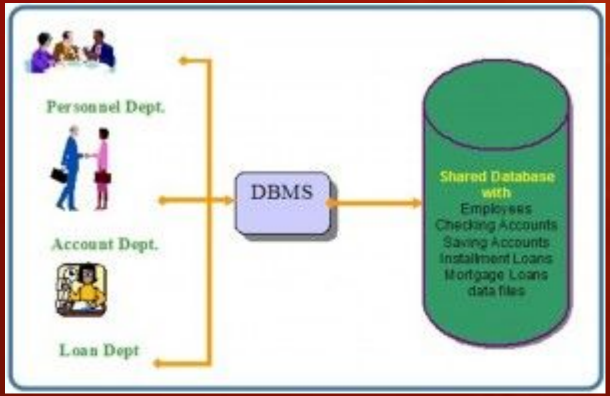
INVENTORY ON ORDER TABLE

Database Software

- Database management system (DBMS)
  - Used to create, maintain, and access databases.

Components of DBMS

Component	Description
Hardware	Physical devices storing data
Software	DBMS programs like MySQL, Access
Data	Actual records in tables
Users	People interacting with the database
Procedures	Rules for data access & management



Database vs File System

File System	Database
Stores files independently	Stores related data together
Manual data linking	Relationships managed automatically
No query facility	SQL-based queries
Difficult to maintain	Easier to manage and secure

Summary

- Database = Organized collection of related data.
- DBMS = Software that manages databases.
- File systems are manual, DBMS is automated and reliable.

Suggested Reading

- Ch-14, Database and Database Management System: Processing and Memory , “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker

- Ch-09, Discovering Computers Fundamentals- Your Interactive Guide -- Gary B Shelly; Misty E Vermaat; Jeffrey J Q